

河北瑞新生物科技有限公司
HEBEI RUIXIN BIOTECHNOLOGY CO., LTD.

Material Safety Data Sheet

VITAMIN B12 FEED ADDITIVE

Section 1 - Chemical Product and Company Identification

MSDS Name: VITAMIN B12 FEED ADDITIVE

Catalog Numbers: CP010A00

Synonyms: Anacobin; Vitamin B12; 1H-Benzimidazole, 5,6-dimethyl-1-(3-O-phosphono-alpha-d-ribofuranosyl)-, monoester with cobinamide cyanide hydroxide, inner salt; Cyanocobalamin.

Trade Name: VITAMIN B12 FEED ADDITIVE

Company Identification:

HEBEI RUIXIN BIOTECHNOLOGY CO., LTD

Section 2 - Composition, Information on Ingredient

Chemical name: Cyanocobalamin

Formular: C63H88CoN14O14P

Formular weight: 1355.38

CAS No.: 68-19-9

Percent: ≥96.0 %

EINECS number: -

Section 3 - Hazards Identification

EMERGENCY OVERVIEW

Caution! May cause eye, skin, and respiratory tract irritation. Light sensitive.

Hygroscopic (absorbs moisture from the air). This is expected to be a low hazard for usual industrial handling.

Target Organs: None.

Potential Health Effects

Eye: Contact may cause transient eye irritation.
Skin: May cause skin irritation. Although allergic reactions to vitamin B12 have generally been attributed to impurities in the preparation, a few patients have reacted positively to

skin testing with purified cyanocobalamin.

Ingestion: May cause gastrointestinal irritation with nausea, vomiting and diarrhea. Ingestion of large amounts may cause convulsions, tremors, muscular contractions or spasticity, diarrhea, or fever.

Inhalation: May cause respiratory tract irritation.

Chronic: Not available. Available evidence suggests that cobalt is not likely to cause cancer in humans except under uncommon or unlikely routes or levels of exposure. Cobalt is an essential element for humans. It forms part of the molecule of cyanocobalamin. The absence of cyanocobalamin is associated with a variety of deficiency-disease states notably dealing with proper erythropoiesis. The lack of vitamin B12 containing cobalt results in anemia.

Section 4 - First Aid Measures

Eyes: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Get medical aid.

Skin: In case of contact, flush skin with plenty of water. Remove contaminated clothing and shoes. Get medical aid if irritation develops and persists. Wash clothing before reuse.

Ingestion: If swallowed, do not induce vomiting unless directed to do so by medical personnel. Never give anything by mouth to an unconscious person. Get medical aid.

Inhalation: If inhaled, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Get medical aid.

Notes to Physician: Treat symptomatically and supportively.

Section 5 - Fire Fighting Measures

General Information: As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. During a fire, irritating and highly toxic gases may be generated by thermal decomposition or combustion.

Extinguishing Media: For small fires, use water spray, dry chemical, carbon dioxide or chemical foam.

Flash Point: Not applicable.

Autoignition Temperature: Not applicable.

Explosion Limits, Lower: Not available.

Upper: Not available.

NFPA Rating: (estimated) Health: 1; Flammability: 1; Instability: 0

Section 6 - Accidental Release Measures

General Information: Use proper personal protective equipment as indicated in Section 8.

Spills/Leaks: Vacuum or sweep up material and place into a suitable disposal container.

Clean up spills immediately, observing precautions in the Protective Equipment section.

Avoid generating dusty conditions. Provide ventilation.

Section 7 - Handling and Storage

Handling: Wash thoroughly after handling. Use with adequate ventilation. Minimize dust generation and accumulation. Avoid contact with eyes, skin, and clothing. Avoid breathing dust.

Storage: Store protected from moisture. Store protected from light. Refrigeration has been recommended

Section 8 - Exposure Controls, Personal Protection

Engineering Controls: Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower. Use adequate ventilation to keep airborne concentrations low.

Exposure Limits

Chemical Name	ACGIH	NIOSH	OSHA - Final PELs
Cyanocobalamin	none listed	none listed	none listed

OSHA Vacated PELs: Cyanocobalamin: No OSHA Vacated PELs are listed for this chemical.

Personal Protective Equipment

Eyes: Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

Skin: Wear appropriate protective gloves to prevent skin exposure.

Clothing: Wear appropriate protective clothing to minimize contact with skin.

Respirators: Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

Section 9 - Physical and Chemical Properties

Physical State: Crystalline powder
Appearance: Light pink, pale red to brown, fine powder, hygroscopicity
Odor: odorless **pH:** aq solns are neutral
Vapor Pressure: Negligible.
Vapor Density: Not available.
Evaporation Rate: Negligible.
Viscosity: Not available.
Boiling Point: > 300 deg C
Freezing/Melting Point: > 300 deg C
Decomposition Temperature: Not available.
Solubility: Moderately soluble in water.
Specific Gravity/Density: Not available.
Molecular Formula: C₆₃H₈₈CoN₁₄O₁₄P
Molecular Weight: 1355.38

Section 10 - Stability and Reactivity

Chemical Stability: Stable under normal temperatures and pressures. May decompose when exposed to light. Darkens at 210 - 220.

Conditions to Avoid: Light, dust generation, excess heat.

Incompatibilities with Other Materials: Strong oxidizers, strong acids.

Hazardous Decomposition Products: Carbon monoxide, oxides of nitrogen, oxides of phosphorus, carbon dioxide, oxides of cobalt.

Hazardous Polymerization: Will not occur.

Section 11 - Toxicological Information

RTECS#:

CAS# 68-19-9: GG3750000

LD50/LC50:

Not available.

Carcinogenicity:

CAS# 68-19-9: Not listed by ACGIH, IARC, NTP, or CA Prop 65.

Epidemiology: No information available.

Teratogenicity: See actual entry in RTECS for complete information.

Reproductive Effects: See actual entry in RTECS for complete information.

Mutagenicity: Please refer to RTECS# GG3750000 for specific information.

Neurotoxicity: No information available.

Other Studies:

Section 12 - Ecological Information

No information

available

Section 13 - Disposal Considerations

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. US EPA guidelines for the classification determination are listed in 40 CFR Parts 261.3. Additionally, waste generators must consult state and local hazardous waste regulations to ensure complete and accurate classification.

RCRA P-Series: None listed.

RCRA U-Series: None listed.

Section 14 - Transport Information

	US DOT	Canada TDG
Shipping Name:	Not regulated as a hazardous material	No information available.
Hazard Class:		
UN Number:		
Packing Group:		

Section 15 - Regulatory Information

US FEDERAL

TSCA

CAS# 68-19-9 is listed on the TSCA inventory.

Health & Safety Reporting List

None of the chemicals are on the Health & Safety Reporting List.

Chemical Test Rules

None of the chemicals in this product are under a Chemical Test Rule.

Section 12b

None of the chemicals are listed under TSCA Section 12b.

TSCA Significant New Use Rule

None of the chemicals in this material have a SNUR under TSCA.

CERCLA Hazardous Substances and corresponding RQs

None of the chemicals in this material have an RQ.

SARA Section 302 Extremely Hazardous Substances

None of the chemicals in this product have a TPQ.

Section 313

This material contains Cyanocobalamin (listed as Cobalt compounds), >96%, (CAS# 68-19-9) which is subject to the reporting requirements of Section 313 of SARA Title III and 40 CFR Part 373.

Clean Air Act:

CAS# 68-19-9 (listed as Cobalt compounds) is listed as a hazardous air pollutant (HAP).

This material does not contain any Class 1 Ozone depleters.

This material does not contain any Class 2 Ozone depleters.

Clean Water Act:

None of the chemicals in this product are listed as Hazardous Substances under the CWA.

None of the chemicals in this product are listed as Priority Pollutants under the CWA.

None of the chemicals in this product are listed as Toxic Pollutants under the CWA.

OSHA:

None of the chemicals in this product are considered highly hazardous by OSHA.

STATE

CAS# 68-19-9 can be found on the following state right to know lists: Pennsylvania, (listed as Cobalt compounds).

California Prop 65

California No Significant Risk Level: None of the chemicals in this product are listed.

European/International Regulations

European Labeling in Accordance with EC Directives

Hazard Symbols:

Not available.

Risk Phrases:

Safety Phrases:

S 24/25 Avoid contact with skin and eyes.

WGK (Water Danger/Protection)

CAS# 68-19-9: No information available.

Canada - DSL/NDSL

CAS# 68-19-9 is listed on Canada's DSL List.

Canada - WHMIS

not available.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the MSDS contains all of the information required by those regulations.

Canadian Ingredient Disclosure List

CAS# 68-19-9 is not listed on the Canadian Ingredient Disclosure List.

Section 16 - Additional Information

MSDS Creation

Date: 07/01/2015

The information above is believed to be accurate and represents the best information currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. Users should make their own investigations to determine the suitability of the information for their particular purposes. In no event shall HEBEI RUIXIN BIOTECHNOLOGY CO., LTD LTD.. be liable for any claims, losses, or damages of any third party or for lost profits or any special, indirect, incidental, consequential or exemplary damages.

ORIGINAL CONTROLLED

河北瑞新生物科技有限公司
HEBEI RUIXIN BIOTECHNOLOGY CO., LTD.

崔少山

Cui Shao Shan



HEBEI RUIXIN BIOTECHNOLOGY CO., LTD.

NO.6 Nangong Section, National Road 308, Luancheng District,
Shijiazhuang, Hebei Province, China

Method of Test

VITAMIN B12 FEED ADDITIVE

This product is got by mixing VitaminB12 with some appropriate carrier, it should contains 90%-130% of labeled content of Vitamin B12.

Description

Cyanocobalamin occurs as dark red crystals or powder. It is hygroscopic.

Identification

A: Transfer about 0.002g of VB12, accurately weighed (accurate to 0.00002g), to a 100-ml volumetric flask, add 80ml of water, and sonicate briefly to dissolve. Dilute with diluent to volume, mix, filtrate to get the test sample (centrifugation if necessary). Take the Ultraviolet Absorption in 1-cm cells, over the spectral range from 300-600nm, the maximum absorbance should appear at $361 \pm 1\text{nm}$ and $550 \pm 2\text{nm}$.

B: Take about 0.002g of VB12, accurately weighed, add 2ml of water, shaking for 10min, centrifugation for 5min, and filtrate to get the test sample.

Take about 0.002g of standard VB12, accurately weighed, add 2ml of water, as the standard solution. Apply the test solution and standard solution $10\mu\text{m}$ respectively, at points about 2.5cm from the lower edge of the plate, use a mixture of ethanol and water (19:1) as the spreader. Place a mark 12cm above the spot point, air-dry the plate. Determine the R_f values for the principal spots, the results should be equal to each other.

Particle size:

The product should 100% pass through No.4 mesh sieve.

Accurately weigh about 20g on an analytical balance and sieve for 3 minutes. All pass the sieve.

Loss on drying

Transfer about 1g (accurate to 0.0002g) of VB12, accurately weighed, to a dried container. Place the loaded bottle in the drying chamber, removing the stopper and leaving it also in the chamber for 2 hours at 105°C . upon opening the chamber, close the bottle promptly, and allow it to come to room temperature in a desiccator before weighing for 30min. The loss on drying should not exceed 5.0%.

The formula is as follow:

$$\text{Loss in dry\%} = \frac{W1-W2}{W} * 100\% \dots\dots\dots(2)$$

In which W1 is the weight before heating, in g, W2 is the weight after heating, in g, W is the weight of the sample. The result is accurately to 0.1.

Assay:

Test solution(a): transfer about 1g of the substance to be examined, accurately weighed, to a 100ml brown volumetric flask, add 60ml deionized water, sonicate to dissolve, and then dilute to the volume. Accurately pipette 1.00ml of the diluted solution into a 50ml brown volumetric flask, dilute

to the volume with deionized water.

Test solution(b): transfer about 0.2g of the substance to be examined, accurately weighed, to a 100ml brown volumetric flask, add 60ml deionized water, sonicate to dissolve, and then dilute to the volume. Accurately pipette 1.00ml of the diluted solution into a 50ml brown volumetric flask, dilute to the volume with deionized water.

The chromatographic procedure may be carried out using:

--- as the stationary phase using C18 column: 4.6mm diameter, 150mm long, particle size 5 μ m.

— as the mobile phase at a flow rate of 0.5ml/min a mixture prepared as follows: mix 1000ml of methanol and water(300+690) , 10ml of glacial acetic acid and 1g of hexane sulfonic acid sodium salt, filter, ultrasonic degassing.

— a spectrophotometer set at 361nm as detector.

— a loop injector.

— the ratio of the retention times is about 7.0min

Calculate the content by the formula:

$$X\% = \frac{P_i * \rho_i * 5000}{P_{at} * m * k * 1000000} \dots\dots\dots(1)$$

In which X% is the content of Cyanocobalamin(%), P_i is the area responses for the major peaks, m is the quality, in g. In the reference solution, ρ_i is the concentration, in μ m per ml, P_{at} is the average area responses for the major peaks, k is the marking content(%). 5000 is the multiple of diluting, 1000000 is the modulus. The result is accurately to 0.1.

Packing and storage

Preserve in tight, light-resistance containers.

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